

Similar Shapes

GCSE Maths · Practice Questions

20 questions · 62 marks total

jtmaths.org · free maths resources

Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1

[1 mark]

Two similar triangles have corresponding sides of 3 cm and 9 cm.
Write down the scale factor of the enlargement.

.....

2

[2 marks]

Rectangle A measures 4 cm by 6 cm. Rectangle B measures 10 cm by 15 cm.
Show that the two rectangles are similar.

.....

3

[2 marks]

Two similar triangles have corresponding sides 5 cm and 15 cm.
Another side of the smaller triangle is 7 cm. Find the corresponding side of the larger triangle.

.....

4

[2 marks]

A shape has perimeter 12 cm. It is enlarged by scale factor 3.
Write down the perimeter of the enlarged shape.

.....

5

[3 marks]

Triangle A has sides 6 cm, 8 cm and 10 cm.
Triangle B is similar to triangle A and its shortest side is 9 cm.
Work out the other two sides of triangle B.

.....

6

[3 marks]

Two triangles are similar. The angles of the first are 40° , 60° and 80° .

- (a) Write down the angles of the second triangle.
 - (b) Explain your answer.
-

7

[3 marks]

A shape has area 20 cm^2 . It is enlarged by scale factor 2.

- (a) Work out the area of the enlarged shape.
 - (b) Explain why the answer is not 40 cm^2 .
-

8

[3 marks]

In triangle ABC, the line DE is parallel to BC, with D on AB and E on AC.

AD = 4 cm, DB = 6 cm and AE = 6 cm.

Work out the length of EC.

9

[4 marks]

A photograph measures 8 cm by 12 cm.

It is enlarged so that the shorter side becomes 20 cm.

- (a) Work out the scale factor.
 - (b) Work out the length of the longer side of the enlargement.
 - (c) Work out the area of the enlargement.
-

10

[4 marks]

In triangle PQR, the line ST is parallel to QR, with S on PQ and T on PR.
PS = 5 cm, SQ = 3 cm and ST = 4 cm.
Work out the length of QR.

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

Two similar shapes have a length scale factor of 4.
Write down the area scale factor.

2

[2 marks]

Two similar solids have a length scale factor of 3.
Write down the volume scale factor.

3

[3 marks]

Two similar shapes have areas 18 cm^2 and 50 cm^2 .
Work out the length scale factor.

4

[3 marks]

Two similar solids have volumes 24 cm^3 and 375 cm^3 .
Work out the length scale factor.

5

[4 marks]

Solid A has surface area 48 cm^2 and volume 32 cm^3 .

Solid B is similar to A and has surface area 108 cm^2 .

Work out the volume of solid B.

6

[4 marks]

Two similar cones have heights 6 cm and 9 cm.

The smaller cone has volume $40\pi \text{ cm}^3$.

Work out the volume of the larger cone, in terms of π .

7

[4 marks]

A cone of height 12 cm has volume 96 cm^3 .

The top of the cone is cut off by a cut parallel to the base, exactly halfway up.

(a) Explain why the small cone removed has volume 12 cm^3 .

(b) Work out the volume of the remaining frustum.

8

[4 marks]

Triangle ABC is similar to triangle PQR.

$AB = 6 \text{ cm}$, $BC = 8 \text{ cm}$ and $PQ = 9 \text{ cm}$. The area of triangle ABC is 24 cm^2 .

(a) Work out the length of QR.

(b) Work out the area of triangle PQR.

9

[4 marks]

Two similar statues are made from the same material.

The smaller statue is 30 cm tall and has mass 4.5 kg. The larger statue is 50 cm tall.

Work out the mass of the larger statue, to 1 decimal place.

10

[5 marks]

In triangle ABC, angle $ABC = 90^\circ$. BD is perpendicular to AC, with D on AC.

AD = 4 cm and DC = 9 cm.

- (a) Explain why triangle ABD is similar to triangle BCD.
 - (b) Work out the length of BD.
 - (c) Work out the length of AB, giving your answer in surd form.
-